



Western Australian Certificate of Education Examination, 2012

Question/Answer Booklet

PHYSICAL EDUCATION STUDIES

Stage 2

Please place your student identification label in this box

Student Number: In figures

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In words

Time allowed for this paper

Reading time before commencing work: ten minutes

Working time for paper: two and a half hours

Materials required/recommended for this paper

To be provided by the supervisor

This Question/Answer Booklet

Multiple-choice Answer Sheet

Number of additional
answer booklets used
(if applicable):

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To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener,
correction tape/fluid, eraser, ruler, highlighters

Special items: non-programmable calculators approved for use in the WACE examinations

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of the examination

The Physical Education Studies examination comprises a written examination worth 70 per cent of the total examination score and a practical examination worth 30 per cent of the total examination score.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Suggested working time (minutes)	Marks available	Percentage of exam
Section One: Multiple-Choice	20	20	30	20	14
Section Two: Short answer	9	9	75	50	35
Section Three: Extended answer	4	2	45	30	21
Total					70

Instructions to candidates

1. The rules for the conduct of WACE external examinations are detailed in the booklet *WACE Examinations Handbook 2012*. Sitting this examination implies that you agree to abide by these rules.
2. Answer the questions according to the following instructions.

Section One: Answer all questions on the separate Multiple-choice Answer Sheet provided. For each question, shade the box to indicate your answer. Use only a blue or black pen to shade the boxes. If you make a mistake, place a cross through that square, then shade your new answer. Do not erase or use correction fluid/tape. Marks will not be deducted for incorrect answers. No marks will be given if more than one answer is completed for any question.

Sections Two and Three: Write answers in this Question/Answer Booklet.

3. You must be careful to confine your responses to the specific questions asked and to follow any instructions that are specific to a particular question.
4. Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.
 - Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
 - Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

Section One: Multiple-choice**14% (20 Marks)**

This section has **20** questions. Answer **all** questions on the separate Multiple-choice Answer Sheet provided.

For each question, shade the box to indicate your answer. Use only a blue or black pen to shade the boxes. If you make a mistake, place a cross through that square then shade your new answer. Do not erase or use correction fluid/tape. Marks will not be deducted for incorrect answers. No marks will be given if more than one answer is completed for any question.

Suggested working time: 30 minutes.

1. In basketball, a player who is fouled while taking a shot is often rewarded a 'free throw', whereby the player is allowed to attempt a shot without any defenders involved. The basketball free throw would be classified as which of the following motor skills?
 - (a) open and fine
 - (b) closed and gross
 - (c) closed and fine
 - (d) open and gross

2. A batter in baseball is watching a pitch and notices the spin of the ball. The batter realises it will be difficult to hit, so does not attempt to swing. According to the information-processing model, what stage of the model did the batter reach?
 - (a) identification of stimuli/input
 - (b) response identification/decision-making
 - (c) response/output
 - (d) feedback

3. Many muscles in the body work in what are called antagonist pairs. The antagonist of the quadriceps muscle group would be the
 - (a) gluteals.
 - (b) Biceps Brachii.
 - (c) hamstrings.
 - (d) Gastrocnemius.

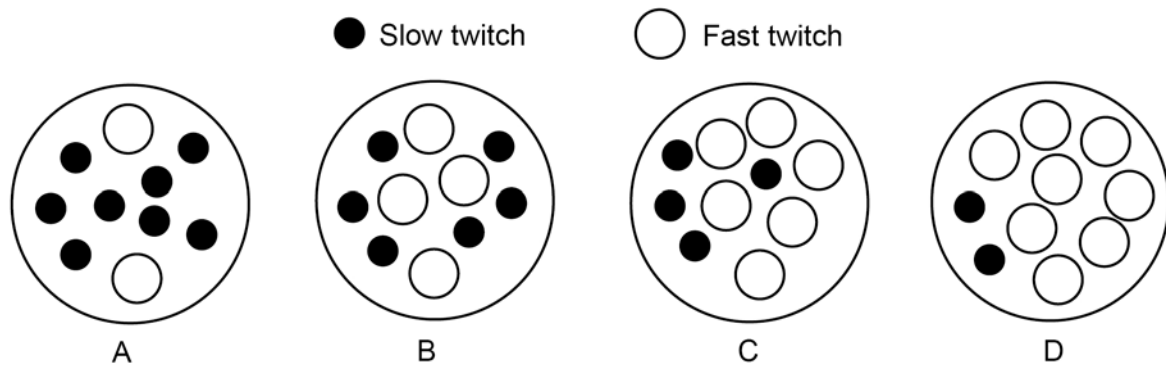
4. Which of the following statements **best** describes Newton's third law of motion?
 - (a) When you push against a wall, the wall pushes back with equal and opposite force.
 - (b) A ball sitting on a tee will remain on the tee until struck by an external force.
 - (c) As a rugby player pushes against another player with more and more force, the player being pushed is accelerated.
 - (d) In lawn bowls, an athlete applies a specific amount of force to allow the ball to reach a specific distance before it stops due to friction.

5. The image below shows a hammer thrower spinning in preparation for releasing the hammer. The hammer is like a 'ball and chain' that the athlete spins with before releasing the hammer for maximum distance. What type of motion **best** describes the hammer in this image?



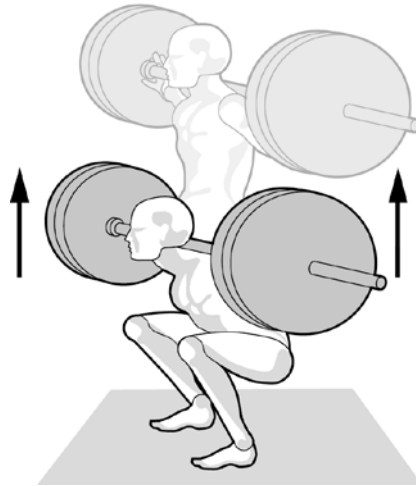
- (a) linear
(b) parabolic
(c) angular
(d) rectilinear
6. When describing the optimum projection angle for a projectile, if it takes off and lands at the same height, in theory, which angle would maximise the horizontal distance it travels?
- (a) 25°
(b) 35°
(c) 45°
(d) 55°
7. Which of the following would decrease during a session of high intensity sprinting?
- (a) venous oxygen volume
(b) heart rate
(c) respiratory rate
(d) arteriovenous oxygen difference
8. Which of the following energy sources would provide the largest amount of energy?
- (a) water
(b) fat
(c) carbohydrate
(d) protein

9. Which of the following muscle samples would **most** benefit an élite sprinter?



- (a) A
(b) B
(c) C
(d) D
10. Blood flow changes are an immediate response to exercise. Which of the following parts of the body would you expect to have decreased blood flow in response to running?
- (a) skeletal muscle
(b) heart
(c) skin
(d) abdominal organs
11. In the sport of rugby, the ball carrier has to react to the defence quickly, often changing his initial running direction. This is characteristic of what component of fitness?
- (a) muscular strength
(b) agility
(c) muscular power
(d) flexibility
12. A figure skater who, after training and while relaxing before sleep, goes through her choreographed routine 'in her head' is practising which mental skill?
- (a) self-talk
(b) self-imagery
(c) relaxation
(d) concentration

13. The image below is an example of the back squat exercise. This exercise is used to maximise strength and ability to produce force. Based on this description, where on the coordination continuum would this exercise most likely be placed?



- (a) simultaneous
(b) sequential
(c) continuous
(d) isometric
14. Two athletes, Scott (15 years old) and Sarah (20 years old) are both élite tennis players for their age group. Which of the following factors would most likely be the reason these athletes have two different levels of arousal to achieve optimal performance?
- (a) gender
(b) age
(c) skill
(d) type of activity
15. Coaches know it is important for their athletes, when trying to accelerate rapidly during short sprints, to be able to apply large amounts of force with each step. This information is descriptive of which of Newton's laws?
- (a) Newton's first law of motion
(b) Newton's second law of motion
(c) Newton's third law of motion
(d) both Newton's first and second laws
16. In soccer, the goalie must respond to 'shots on goal' as quickly as possible. Although not ideal, a goalie with lower levels of speed or agility may still be successful if they possess an exceptional level of which of the following components of fitness?
- (a) cardiorespiratory endurance
(b) coordination
(c) flexibility
(d) reaction time

17. Which chamber of the heart is responsible for pumping blood through the aorta for distribution to the rest of the body?
- (a) left ventricle
 - (b) left atrium
 - (c) right ventricle
 - (d) right atrium
18. Which bones are located in both the toes and fingers?
- (a) metacarpals
 - (b) metatarsals
 - (c) phalanges
 - (d) carpals
19. In rehabilitation, a resistance training machine that maintains constant velocity throughout the entire exercise is used to test, diagnose and train individuals recovering from injury. This machine is using which resistance training type?
- (a) isometric
 - (b) isotonic
 - (c) isokinetic
 - (d) isospecific
20. An adaptation of long-term aerobic training is improved utilisation of which of the following sources of energy during submaximal exercise?
- (a) fat
 - (b) carbohydrate
 - (c) protein
 - (d) water

End of Section One

See next page

Section Two: Short answer

35% (50 Marks)

This section has **nine (9)** questions. Answer **all** questions. Write your answers in the spaces provided in this Question/Answer Booklet. Wherever possible, confine your answers to the lines provided. Use a blue or black pen (**not** pencil) for this section.

Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

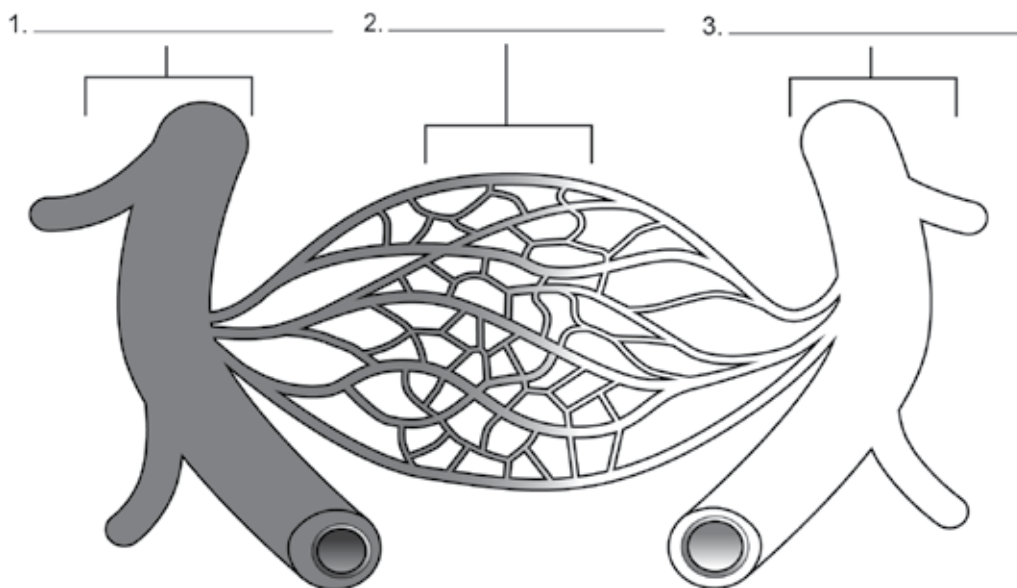
- Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
- Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

Suggested working time: 75 minutes.

Question 21

(5 marks)

The following diagram represents the location where gas is exchanged in the circulatory system. The grey side represents de-oxygenated blood and the white side represents oxygenated blood.



(a) Insert labels for the **three** structures indicated in the above diagram. (3 marks)

(b) Describe how gas exchange occurs at the alveoli. (2 marks)

See next page

Question 22**(6 marks)**

Skeletal muscle tissue has four major characteristics. One of these characteristics is excitability. Excitability describes muscle tissue's ability to respond to a stimulus. Name and describe the other **three** characteristics of muscle tissue.

Characteristic one: _____

Description one: _____

Characteristic two: _____

Description two: _____

Characteristic three: _____

Description three: _____

Question 23

(6 marks)

Using the images below:

- (a) Identify with an 'X' where each activity fits on the associated continuum of skill classification. (3 marks)
- (b) Justify the reason for the position on the continuum of skill classification you chose for each skill. (3 marks)



Figure 1: Batter

Open ←————→ Closed

Justification: _____



Figure 2: Runner

Discrete ←————→ Continuous

Justification: _____



Figure 3: Pitcher

Fine ←————→ Gross

Justification: _____

Question 24

(6 marks)

Below is a picture of Sally Pearson who competes for Australia in the 100 metre hurdles. Her personal best time is 12.28 seconds. She is exceptionally skilful in not letting her trail leg hit the hurdles as she passes over them. Using the image, complete the following table in relation to the trail leg as pictured.



Joint	Articulating bones	Movement
Hip	1. _____	
	_____	_____
	_____	_____
	2. _____	_____

Knee	1. _____	
	_____	_____
	_____	_____
	2. _____	_____

See next page

Question 25

(7 marks)

Caitlin Basset is a member of the West Coast Fever Netball team. As an athlete, she is in the Autonomous Stage of the Fitts and Posner model of motor learning. During a game of netball, she is fluent in her execution of skills.



- (a) Describe **three** other characteristics of this stage. (3 marks)

One: _____

Two: _____

Three: _____

- (b) Muscular endurance and lower body muscular power are important fitness components for Caitlin. Define these components and name a type of training that would improve each fitness component. (4 marks)

Muscular endurance: _____

Type of training: _____

Muscular power: _____

Type of training: _____

See next page

Question 26**(5 marks)**

In the space below, draw a graph (labelling each axis) that represents the potential relative energy contribution and timing of the three energy systems for a running event that lasts approximately 5 minutes.

**See next page**

Question 27

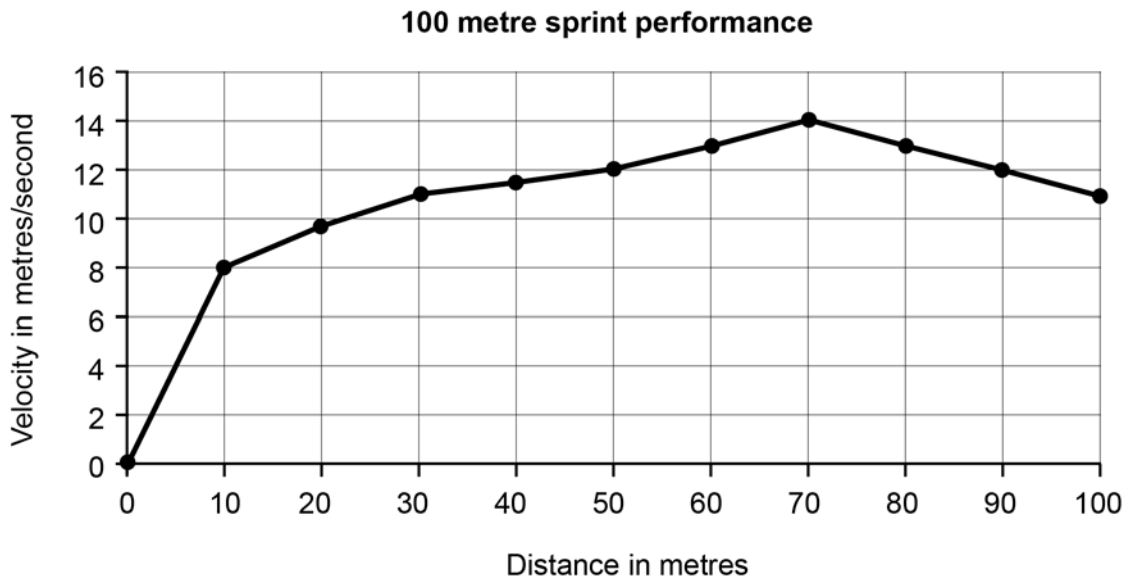
(5 marks)

In an Olympic year, the 100 metre sprint is of great interest to many people, as it 'crowns the fastest human'. Use the diagram below to help you understand how one of these runners completes the race.

- (a) Is the average velocity or peak instantaneous velocity higher for a 100 metre runner? (1 mark)

- (b) Label on the graph where you would find:

- (1) the largest acceleration and
(2) the highest instantaneous velocity. (2 marks)



- (c) Justify the location you selected for (1), the largest acceleration and (2), the highest instantaneous velocity. (2 marks)

Question 28

(4 marks)

Nideffer's model identifies four main attentional styles. Write the attentional style required for each of the following situations.

Style	Situation
	Bowler deciding what type of spin to place on his next delivery
	A tennis player focusing on returning a serve
	A golfer mentally rehearsing his drive from the tee
	Attacker dribbling ball down field checking positions of opposition players

Question 29**(6 marks)**

On the diagram below, name and label the following muscles:

- (a) The muscle known as the most powerful hip extensor. (2 marks)

- (b) The muscle responsible for dorsiflexion. (2 marks)

- (c) The muscles responsible for elbow flexion. (2 marks)

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End of Section Two

See next page

Section Three: Extended answer**21% (30 Marks)**

This section contains **four (4)** questions. You must answer **two (2)** questions. Write your answers in the spaces provided.

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- Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

Suggested working time: 45 minutes.

Question 30**(15 marks)**

Aaron is competing in the opening round of the shot put competition at the Olympic Games. His first attempt resulted in a performance well below his usual standard. After his first attempt, Aaron said to himself that he “felt as if he didn’t put enough force into the throw” as he saw the distance up on the scoreboard, and at the same time, his coach told him he needed to “release the shot in an extended position with a higher right shoulder”. The images below represent the path of the shot put during his first and second attempts.

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- (a) Name and describe the **three** components of projectile motion that influenced his results. (6 marks)

See next page

[illegible]

- (b) Name and describe **three** types of feedback that were provided by the coach and Aaron during his performance. (6 marks)

[illegible]

- (c) Using specific examples from the scenario on page 18, justify how each type of feedback received by Aaron related to the changes in performance that occurred between his first and second attempts. (3 marks)

Question 31

(15 marks)

In 2011, Australian Craig Alexander won his third Hawaiian Ironman in the record time of 8 hours 3 minutes and 56 seconds. This race requires athletes to swim 3.8 km, cycle 180 km and then run a marathon (42 km).

- (a) Identify and describe **five** principles of training for continued success in the Hawaiian Ironman event. (10 marks)

[illegible]

- (b) Part of Craig's race plan is to maintain energy stores to reduce the onset of fatigue. If Craig fails to follow his supplementation plan he will be unsuccessful in his race. Explain, from the onset of his race, how his body will utilise each energy source through the race and the physiological consequence of only drinking water during the race. (5 marks)

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

Question 32

(15 marks)

Occasionally swimmers (such as Jenny in the image below) lose balance while on the starting block, causing them to make a false start.

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but may be viewed at
www.theage.com.au/articles/2004/03/30/1080544485954.html..

- (a) Demonstrate your understanding of the biomechanical principle of balance by outlining **three** factors that influence stability while on the starting blocks and describe how each factor could be modified to improve Jenny's stability. (6 marks)

[illegible]

See next page

- (b) Jenny is participating in her first senior competition. This is causing her anxiety during her performance. Demonstrate your knowledge of the inverted U hypothesis by drawing a labelled diagram below. Include an arousal curve for Jenny and a second arousal curve that represents an elite swimmer. Indicate where an ideal arousal level is for Jenny on her curve and where her actual arousal level was for the performance demonstrated by the image on page 23. (6 marks)



- (c) Describe how age, skill level and type of activity can influence arousal regulation.

(3 marks)

Question 33

(15 marks)

Novak Djokovic won the Australian Open Tennis championship in 2012. His success was not only a result of his skilful execution of his shots, but also the efficiency of his cardiorespiratory system.

- (a) Describe **three** immediate responses of Novak's respiratory system during a game of tennis. (3 marks)

- (b) From his time as a junior tennis player to his Australian Open win, describe **six** long-term adaptations of the circulatory system that would have occurred to Novak from his years of training. (6 marks)

- (c) Name the **two** stages, and describe **two** characteristics of each stage, of the Fitts and Posner model of motor learning that Novak would have passed through successfully during his junior tennis years before he became a professional. (6 marks)

End of questions

Question number: _____

[illegible]

Question number: _____

[illegible]

Question number: _____

[illegible]

Spare Answer page

Question number: _____

[illegible]

Question number: _____

[illegible]

ACKNOWLEDGEMENTS

Section One

Question 5 Photograph adapted from: *From 1987 to 2007 Tikhon inherits Litvinov's mantle*. (2007, August 27). Monaco: International Association of Athletics Federations. Retrieved April 14, 2012, from <http://osaka2007.iaaf.org/>. Copyright Getty Images.

Question 9 Diagram of muscle samples by courtesy of the examining panel.

Question 13 Adapted from: Rexwar. (2007). Deep squat [Image]. *Wikipedia*. Retrieved April 14, 2012, from <http://en.wikipedia.org/wiki/File:Squat.png>. Licensed under the Creative Commons Attribution 3.0 Unported license.

Section Two

Question 21 Abhyasika. (n.d.). *Gas exchange in the circulatory system* [Image]. Retrieved April 14, 2012, from www.abhyasika.com/10/10EM/10EMSandT/10EMSandT10/CapillaryNetwork.JPG.

Question 23b *Cricketer* [Image]. (n.d.). Retrieved April 14, 2012, from www.middlepeg.com/images/sachi.jpg.

Lightning Bolt [Image] from: Olympic hero Usain Bolt set for summer sprint at Crystal Palace. (2009, April 20). *Daily Mail*. Retrieved April 14, 2012, from www.dailymail.co.uk/sport/othersports/article-1172050/. Copyright Reuters.

Baseball pitcher [Image] from: *Baseball – The importance of pitching confidence*. (n.d.). Retrieved April 14, 2012, from www.yupedia.com/baseball-the-importance-of-pitching-confidence.html.

Question 24 Adapted image Shortlisted from: Stellar season sees Pearson leap into award contention. (2011, October 4). *Brisbane Times*. Retrieved April 14, 2012, from www.brisbanetimes.com.au/sport/. Copyright Getty Images.

Question 25 Paddynapper. (2008). Caitlyn Bassett [Image]. *Wikipedia*. Retrieved April 14, 2012, from http://en.wikipedia.org/wiki/File:Caitlin_Bassett.jpg. Licensed under the Creative Commons Attribution-Share Alike 2.0 Generic license.

Question 27 Graph of 100 metre sprint performance by courtesy of the examining panel.

Question 29 *Muscle anatomy* [Image]. (n.d.). Retrieved April 14, 2012, from www.shapesense.com/images/blank-muscle-anatomy.jpg.

Section Three

Question 30 Adapted from: Walter, P. (2011, December 4). *Jacko Gill* [Image]. Retrieved April 14, 2012, from www.zimbio.com/pictures/0cGu2qPW5IL/2011+Shot+Put+Invitational+SerSer/rMjmaxEshq. Copyright Phil Walter/Getty Images AsiaPac.

Question 32 Clayton, T. (n.d.). Swimmers. [Image]. *The Age*. Retrieved April 14, 2012, from www.theage.com.au/articles/2004/03/30/1080544485954.html.

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